

Real-Time Multilingual Sign Language Recognition

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ABSTRACT

Automated Sign Language Recognition (SLR) systems offer a vital bridge between hearing and deaf communities, yet current frameworks remain **predominantly monolingual** and require high-specification hardware. This research introduces a **multilingual, real-time SLR desktop application** capable of classifying American Sign Language (ASL), British Sign Language (BSL), and a novel third category of cross-linguistically equivalent **MUTUAL signs** — all running on consumer-grade hardware without a GPU.

A custom self-collected dataset of **2,726 landmark sequences** across 109 sign classes was recorded using MediaPipe Hands. A stacked LSTM network processes 45-frame, 126-feature skeletal trajectories. Evaluated on a 546-sequence held-out test set, the system achieves **93.77% overall accuracy** and a macro F1 of 93.52%, with inference below 33 ms per frame.

RESEARCH QUESTIONS

- RQ1** Can a single landmark-based LSTM model reliably distinguish ASL, BSL, and MUTUAL signs within a low latency, real-time framework on consumer hardware?
- RQ2** Does treating cross-linguistically equivalent signs as a unified MUTUAL class produce superior classification results compared to maintaining separate instances per given language?
- RQ3** How significantly does data augmentation improve model generalisation within a training set with an individual signer?

MOTIVATION & BACKGROUND

Sign languages are **natural languages** with distinct grammars and phonological systems. Despite sharing roots culturally, ASL and BSL are mutually unintelligible, common concepts such as "like" or "doctor" are signed entirely differently across the two languages.

Contemporary SLR systems are almost exclusively monolingual and require high specification hardware. This project addresses that gap with a deployable, without the requirement of GPU.

NOVEL CONTRIBUTION: MUTUAL CLASS

Signs that are **phonologically equivalent** across ASL and BSL are treated as a single *MUTUAL* class rather than separate per-language entries. This amalgamates training samples from both language communities, optimising learning efficiency under single-signer dataset constraints. To the author's knowledge, this constitutes a **novel contribution** to the multilingual SLR design space.



HARDWARE & DEPLOYMENT

CPU	Intel Core i5-1035G7 (1.50 GHz)
RAM	8 GB
GPU	Intel Iris Plus (integrated)
Camera	Standard 1080p built-in webcam
Network	No network dependency

The hardware constraint was deliberate, for the purpose of mirroring the intended deployment environment.

SYSTEM ARCHITECTURE

STEP 1 Data Collection Isolated ASL / BSL, Mutual signs, Session-based splits, Augmentation	STEP 2 Feature Extraction MediaPipe Hands, 126 skeletal landmarks, Coordinate flattening per frame	STEP 3 Model Training Stacked LSTM, 109 sign classes, Baseline vs. augmented runs	STEP 4 Real-Time Inference Local desktop UI, Rolling 5-frame window, Zero network dependency
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The architecture extracts **126-dimensional skeletal landmark sequences** (21 keypoints × 3 coordinates × 2 hands) from a standard webcam, classified by a stacked LSTM across 109 sign classes.

MODEL ARCHITECTURE (STACKED LSTM)

LAYER	DETAIL
Input	45 × 126
Masking	mask_value = 0.0
LSTM (return sequences)	128 units
Dropout	p = 0.30
LSTM	64 units
Dropout	p = 0.30
Dense (ReLU)	64 units
Dense (Softmax)	109 classes

DATASET STRUCTURE

LANGUAGE	CLASSES	TRAIN SEQS	TEST SEQS	REC./SIGN
ASL	50	1,000	250	20 / 5
BSL	49	980	245	20 / 5
MUTUAL	10	200	51	20 / 5
Total	109	2,180	546	—

Sequences: 45 frames, 126 features. Split: 80/20 train/validation stratified. Aug-ON applies Gaussian noise ($\sigma=0.01$), random scaling (0.9–1.1×) and temporal warping (0.85–1.15×) at training time.

TRAINING & VALIDATION PERFORMANCE

100.00% AUG-ON · BEST VAL ACC (EPOCH 50)	98.996% AUG-OFF · BEST VAL ACC (EPOCH 60)
1.2×10 ⁻⁵ AUG-ON · FINAL VAL LOSS	0.0401 AUG-OFF · FINAL VAL LOSS

The augmented model trained for the full 100 epoch budget and validation accuracy remained at 100% from epoch 50 onwards. Aug-OFF employed early stopping (patience 15), terminating at epoch 72.

OVERALL TEST RESULTS (AUG-ON)

93.77% OVERALL ACCURACY	93.52% MACRO F1 SCORE
95.52% PRECISION (MACRO)	93.76% RECALL (MACRO)

512 / 546 test sequences correctly classified

PER-LANGUAGE ACCURACY (AUG-ON)

ASL	92.40%
BSL	94.69%
MUTUAL	96.08%

Key finding: MUTUAL achieves the highest per-language accuracy at 96.08%, validating the data amalgamation design decision. Only 2 errors occur, DRINK misclassified as PHONE on 2 of 5 test samples.

AUGMENTATION IMPACT (AUG-ON VS. AUG-OFF)

METRIC	AUG-ON	AUG-OFF	Δ
Overall Accuracy	93.77%	86.45%	+7.32 pp
Precision (macro)	95.52%	87.84%	+7.68 pp
Recall (macro)	93.76%	86.42%	+7.34 pp
F1 Score (macro)	93.52%	84.94%	+8.58 pp
Total Errors	34	74	−54.1%

Data augmentation **halves the error count** from 74 to 34, confirming that synthetic spatial and temporal variations substantially improve generalisation.

ERROR ANALYSIS

ASL cluster: chair, corn, candy, sit suffer from persistent confusions from near-identical hand to cheek dynamic motions at the wrist and fingertip trajectory level.

BSL cluster: house/stop and close/slow are phonologically similar hand orientations misaligned in landmark space.

Cross-language: 16 of 34 errors (47.1%) involve cross-linguistic confusion. ASL_fast as BSL_good, and ASL_know as BSL_before reflect shared hand orientations rather than semantic similarity.

REAL-TIME PERFORMANCE

<33 ms MAX LATENCY / FRAME	30 FPS SUSTAINED FRAME RATE
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45 FRAMES PER SEQUENCE	5 SMOOTHING WINDOW
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A rolling 5 frame majority smoothing window suppresses flickering; a sign is displayed only when it constitutes the majority class within the window with softmax confidence ≥ 0.20. A 0.5s settle delay prevents premature predictions.

CONCLUSIONS

- Cross-linguistic SLR across ASL, BSL, and MUTUAL on consumer-grade hardware is demonstrably feasible at 93.77% accuracy.
- The MUTUAL class achieves 96.08%, the highest of all three groups, validating data amalgamation as an effective design strategy.
- Data augmentation reduces errors by 54.1% (74 to 34), providing consistent benefit across the 109 class vocabulary.
- Real-time inference maintains <33 ms latency at 30 FPS without network dependency.
- The single-signer constraint limits generalization. Cross-signer performance requires future multi-signer data

FUTURE WORK

- Multi-signer datasets or transfer learning from large-scale corpora (BSL-1K, WLASL) to improve cross-signer generalisation.
- MediaPipe Holistic for body and facial landmark enrichment capturing non-manual features that distinguish phonologically similar pairs.
- Temporal attention mechanisms to better resolve subtle spatial features in short 45 frame sequences.
- Systematic cross-linguistic phonological analysis to identify further MUTUAL equivalence categories.